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Research on indoor multi-sensor UAV altitude fusion and fault-tolerant simulation based on adaptive Kalman filter

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Abstract. Unmanned aerial vehicles (UAVs) are increasingly used in civilian and military applications, where accurate altitude estimation is critical for safe and reliable flight. In GPS-denied indoor environments, single sensors such as barometers or laser rangefinders often suffer from noise, drift, and sudden measurement faults, limiting their reliability. This paper proposes a multi-sensor altitude fusion algorithm based on an adaptive Kalman filter that integrates data from a barometer, laser rangefinder, and accelerometer. A barometer bias term is included in the state-space model for continuous calibration, while a residual-driven adaptive mechanism dynamically tunes process and measurement noise to suppress laser anomalies. Simulation results show that the proposed method improves altitude estimation accuracy and maintains robustness during sensor disturbances, making it suitable for UAV indoor navigation and control tasks.

1. Introduction

In recent years, unmanned aerial vehicles (UAVs) have been widely applied in fields such as agriculture, logistics, aerial photography, environmental monitoring, and scientific research. Accurate altitude measurement is essential for safe flight, as it directly affects navigation, positioning, and control performance. For outdoor UAVs, common sensors include the Global Navigation Satellite System (GNSS, e.g., GPS or BDS), inertial measurement units (IMU), and barometers. While GNSS offers high accuracy at low cost, it becomes unavailable in indoor environments. Barometers, though simple, suffer from noise and long-term drift, whereas IMUs provide precise short-term estimates but accumulate errors over time. Indoor UAV operation is particularly challenging due to the absence of GNSS signals, limited space, and complex interference, which make single-sensor solutions unreliable. To overcome these limitations, multi-sensor fusion has been increasingly adopted to combine complementary measurements from different sensors, improving altitude estimation accuracy, robustness, and reliability [3,4]. However, traditional Kalman filter-based fusion methods [5] often assume stable measurement conditions and lack adaptive fault-tolerant mechanisms.

Pritzl et al. proposed an adaptive multi-sensor fusion approach to address the challenges of UAV altitude estimation in complex indoor environments, including insufficient geometric features, uneven ground, and sensor measurement delays [1]. Their method fuses measurements from a 1D laser rangefinder, a 3D LiDAR SLAM algorithm, a barometer, an IMU, and the UAV's position controller. Simulation results demonstrated that the approach maintains high robustness and accuracy even under degraded or delayed measurements, supporting stable UAV flight in complex indoor scenarios. Geng and Chu Lin developed a navigation scheme based on multi-height sensor fusion and fault-tolerant Kalman filtering (FTKF) [2]. They fused data from barometers, radar altimeters, and laser rangefinders,



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and applied residual chi-square tests and state chi-square tests to detect and isolate faulty sensors. Even when a single sensor failed, the method relied on other sensors to maintain stable system operation. Simulation results showed that the method demonstrated strong reliability and robustness in UAV altitude fusion and fault-tolerant control.

Yu et al. proposed a multi-sensor altitude prediction algorithm based on Kalman filtering [3], which integrates accelerometer, barometer, and laser rangefinder data for UAV altitude estimation. It can compensate for a single sensor failure using data from other sensors, exhibiting strong fault tolerance. Experimental results showed that the algorithm effectively reduced noise, improved altitude prediction accuracy, and maintained stability and robustness in GPS-denied indoor environments.

Long et al. addressed the problem of altitude control instability in hexacopter UAVs caused by altitude estimation bias and varying payload mass [6]. A height measurement system based on barometer–accelerometer fusion via Kalman filtering was developed, along with a dual-loop PID controller enhanced with feedforward control. This approach was applied to a hexacopter plant protection UAV and achieved accurate altitude estimation and stable altitude control.

Therefore, Kalman filter–based multi-sensor altitude fusion is an effective way to improve UAV altitude-hold performance. Since GPS is unavailable indoors, this paper focuses on fusing barometer, laser rangefinder, and accelerometer data. To address sudden laser disturbances and slow barometer drift, a fault-tolerant adaptive Kalman filter is designed with an augmented state model that introduces a barometric bias term for continuous calibration. Through adaptive noise covariance adjustment, the method maintains stability and accuracy even during laser anomalies.

2. Multi-Sensor Observation Mode

Due to the lack of GPS signals in indoor environments, this study focuses on alternative sensors such as barometers, laser rangefinders, and accelerometers for altitude estimation, without relying on GPS. Moreover, this paper does not actually conduct unmanned aerial vehicle (UAV) flights. Instead, it studies the highly integrated algorithm of UAVs indoors in MATLAB.

2.1 Barometer Model

The barometer estimates altitude from atmospheric pressure, temperature, and elevation. The relation is [7]:

$$H = \frac{T_b}{\beta} \left[\left(\frac{p_H}{p_b} \right)^{\frac{\beta R}{g_n}} - 1 \right] + H_b \quad (\beta \neq 0) \quad (1)$$

where H is altitude, T_b the base temperature, β the lapse rate, p_H the measured pressure, p_b the base pressure, R the gas constant, g_n gravity, and H_b the base geopotential height. Since the proposed altitude fusion algorithm is designed for indoor environments, the ideal barometer measurements in this study are calculated using a simplified International Standard Atmosphere (ISA) model. It is assumed that the indoor temperature remains constant at $T_0 = 25^\circ\text{C}$. In the troposphere, the temperature decreases linearly with altitude H , and the atmospheric pressure decreases exponentially with altitude. The ideal pressure at a given altitude H can therefore be expressed as:

$$P(H) = P_0 \times 10^{\left[\frac{5.256 \ln \left(1 - \frac{h}{153.8462 T_0} \right)}{1} \right]} \quad (2)$$

where $P_0 = 101325$ Pa is the standard sea-level atmospheric pressure. The corresponding altitude can be calculated from the measured pressure using:

$$h = - \frac{\left(\left(\frac{P}{P_0} \right)^{1/5.257} - 1 \right) \cdot (T + 273.15)}{0.0065} \quad (3)$$

where $T = 25 - 0.0065H$ represents the temperature corrected for altitude. Considering that barometer measurements are affected by both random noise and slow drift, the model can be expressed as:

$$P_n(k) = P(k) + n_p(k) + d_p(k) \quad (4)$$

where $n_p(k)$ is zero-mean Gaussian white noise representing measurement noise, and $d_p(k)$ is a slowly varying drift term representing the long-term bias of the barometer. The final pressure $P_n(k)$ is converted to the barometric altitude output by applying (3):

$$z_b(k) = f(P_n(k)) = h(k) + e(k) \quad (5)$$

where $h(k)$ is the true altitude, and $e(k)$ is the altitude deviation.

2.2 Laser Rangefinder Model

The laser rangefinder measures distance by emitting a narrow laser beam and calculating the time between emission and reception of the reflected signal. It is lightweight, compact, easy to use, and offers fast, accurate measurements with errors significantly smaller than those of conventional optical rangefinders. Its measurement model can be expressed as:

$$z_l(k) = h(k) + n_l(k) \quad (6)$$

where $z_l(k)$ denotes the measurement output of the laser rangefinder, and $n_l(k)$ is zero-mean Gaussian white noise with variance σ_l^2 . It should be noted that in complex indoor environments, laser range measurements may suffer from step-like disturbances caused by variations in reflective surfaces or temporary occlusions.

2.3 Accelerometer Model

Assuming that the accelerometer error follows Gaussian white noise, the measurement equation of the accelerometer can be expressed as^[8]:

$$z_a(k) = a(k) + n_a(k) \quad (7)$$

where $z_a(k)$ is the vertical acceleration measurement output, $a(k)$ is the true vertical acceleration of the UAV, and $n_a(k)$ is the accelerometer measurement noise. The navigation frame (ENU, origin at UAV center of mass), and body frame (fixed to the UAV) are defined as in Figure 1. The accelerometer measures the three-axis acceleration in the body frame, denoted as $a_b = [a_{bx}, a_{by}, a_{bz}]^T$, which must be transformed into the navigation frame $a_g = [a_{gx}, a_{gy}, a_{gz}]^T$ and corrected for gravity. This transformation can be achieved using a direction cosine matrix (DCM) C_g^b , and the relationship is:

$$a_g = [a_{gx}, a_{gy}, a_{gz}]^T = C_g^b \cdot [a_{bx}, a_{by}, a_{bz}]^T \quad (8)$$

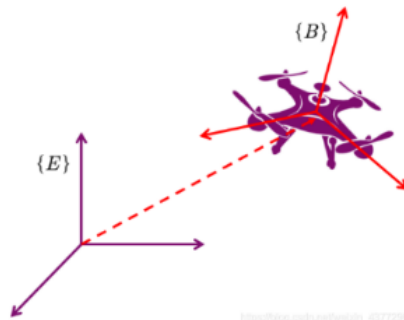


Figure 1. The relationship between the navigation coordinate system and the body coordinate system.

The transformation is typically parameterized using the attitude Euler angles-roll, pitch, and yaw—denoted as ϕ , θ , and ψ , respectively. DCM C_g^b can be expressed as:

$$C_g^b = \begin{bmatrix} \cos \psi \cos \theta & \cos \psi \sin \theta \sin \phi - \sin \psi \cos \phi & \cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi \\ \sin \psi \cos \theta & \sin \psi \sin \theta \sin \phi + \cos \psi \cos \phi & \sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi \\ -\sin \theta & -\sin \theta & \cos \theta \cos \phi \end{bmatrix} \quad (9)$$

Let a_{gz} represent the vertical component of the UAV acceleration in the navigation frame. Then, the UAV's acceleration relative to the Earth can be expressed as:

$$a = a_{gz} - g \quad (10)$$

3. System Model and Algorithm Principle

3.1 System Model Establishment

To achieve accurate UAV altitude estimation and continuous calibration of the barometer bias, this paper selects altitude h , vertical velocity v , and barometer bias e as state variables. The state vector is defined as $x(k) = [h(k), v(k), e(k)]^T$. The system input is the vertical acceleration $u(k) = a$. The measurement vector is $z(k) = [z_b(k), z_l(k)]^T$, where $z_b(k)$ is the barometer measurement, and $z_l(k)$ is the laser rangefinder measurement. The discrete-time state equation of the Kalman filter can be expressed as:

$$x(k+1) = Ax(k) + Bu(k) + w(k-1) \quad (11)$$

The state transition matrix is $A = \begin{bmatrix} 1 & T & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, the input matrix is $B = \begin{bmatrix} \frac{T^2}{2} & T & 0 \end{bmatrix}^T$, and T is

the sampling period. $w(k-1)$ denotes the process noise, which is assumed to follow a zero-mean Gaussian distribution with covariance $Q = \text{diag}(\sigma_h^2, \sigma_v^2, \sigma_e^2)$. The measurement equation is given by:

$$z(k) = Hx(k) + v(k) \quad (12)$$

where $H = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ is the observation matrix, and $v(k)$ represents the measurement noise with zero mean and covariance $R = \text{diag}(\sigma_b^2, \sigma_l^2)$.

3.2 Kalman Filtering and Adaptive Noise Adjustment

Based on the established state and measurement equations, this paper adopts an extended Adaptive Kalman Filter (AKF) to achieve multi-sensor altitude fusion and fault-tolerant estimation. The basic recursive process of Kalman filtering consists of two steps: prediction and update.

The prediction step can be expressed as:

$$\hat{x}(k|k-1) = A\hat{x}(k-1|k-1) + Bu(k) \quad (13)$$

$$P(k|k-1) = AP(k-1|k-1)A^T + Q \quad (14)$$

where $\hat{x}(k|k-1)$ denotes the one-step predicted state estimate, and $P(k|k-1)$ is the predicted error covariance matrix.

The update step involves the following operations:

The innovation covariance is given by Equation (15):

$$r(k) = z(k) - H\hat{x}(k|k-1) \quad (15)$$

The Kalman gain is computed using Equation (16):

$$K(k) = P(k | k-1)H^T [HP(k | k-1)H^T + R(k)]^{-1} \quad (16)$$

The state and error covariance are updated according to Equations (17) and (18):

$$\hat{x}(k | k) = \hat{x}(k | k-1) + K(k)r(k) \quad (17)$$

$$P(k | k) = [I - K(k)H]P(k | k-1) \quad (18)$$

3.2.1 Adaptive Adjustment of Measurement Noise Covariance R . Because laser rangefinders in indoor environments are susceptible to sudden disturbances, a fixed measurement noise covariance matrix RRR cannot accurately capture real measurement uncertainty. In this work, RRR is adaptively tuned based on residual statistics, as expressed in Equation (19):

$$R(k) = \alpha R(k-1) + (1-\alpha)[r(k)r(k)^T + HP(k | k-1)H^T] \quad (19)$$

where $\alpha \in (0,1)$ is a smoothing factor used to prevent over-adjustment caused by a single abrupt disturbance. When the laser measurement residual $r_l(k)$ exceeds a preset threshold, the corresponding R_{laser} is further increased. This reduces the influence of laser measurements on the altitude estimation, thereby increasing the filter's reliance on barometer readings and improving the fault tolerance of the system.

3.2.2 Adaptive Adjustment of Process Noise Covariance Q . In altitude state estimation, the process noise covariance Q reflects the model's tolerance for free state variations. In this study, Q is adaptively adjusted based on the correlation between the Kalman gain and the residual, as expressed in Equation (20):

$$Q(k) = \alpha Q(k-1) + (1-\alpha)K(k)r(k)r(k)^T K(k)^T \quad (20)$$

When the system operates under steady conditions, Q gradually decreases, resulting in smoother altitude estimation. Conversely, during periods of laser measurement disturbance or significant barometric drift, Q is increased to enhance the model's adaptability to abnormal conditions. By jointly adapting both R and Q , the Kalman filter can automatically reduce the weight of laser measurements during abnormal events, while relying more on the barometer's stable trend to maintain continuity of altitude estimation. Furthermore, the inclusion of the barometric bias state allows gradual calibration of accumulated barometer errors, thereby improving the overall estimation accuracy and robustness of the system.

4. Simulation Results and Analysis

4.1 Simulation Environment and Parameter Settings

To verify the effectiveness of the proposed adaptive Kalman filter-based multi-sensor altitude fusion and fault-tolerant method, a MATLAB simulation platform was established. The true UAV altitude was set to a constant value of $15m$. The sampling period was $T = 0.1s$, and the total number of simulation steps was $N = 6000$. The barometer measurement was modeled as the ideal pressure with added zero-mean Gaussian noise ($\sigma_{np} = 1Pa$) and a linear drift increasing from 0 to 3 Pa over the simulation. The laser rangefinder measurements were modeled as Gaussian, with a mean equal to the true altitude and standard deviation $\sigma_{laser} = 0.01m$. To simulate the situation where an obstacle appears below the UAV, a disturbance was introduced to the laser rangefinder measurements during $t = 300$ to $400s$. The initial filter state was set to $h = 14.5m$, $v = 0$, $a = 0$, and $e = 0$.

4.2 Altitude Estimation Results

The following figures illustrate two scenarios of sudden disturbances in the laser rangefinder measurement when simulating the presence of an obstacle beneath the UAV, along with the true altitude,

barometer measurement, laser measurement, and altitude estimation results under different algorithms. The two cases correspond to a $-1m$ step change and a ramp disturbance of the laser measurement occurring between $t = 300$ and $t = 400s$, as shown in Figure 2.

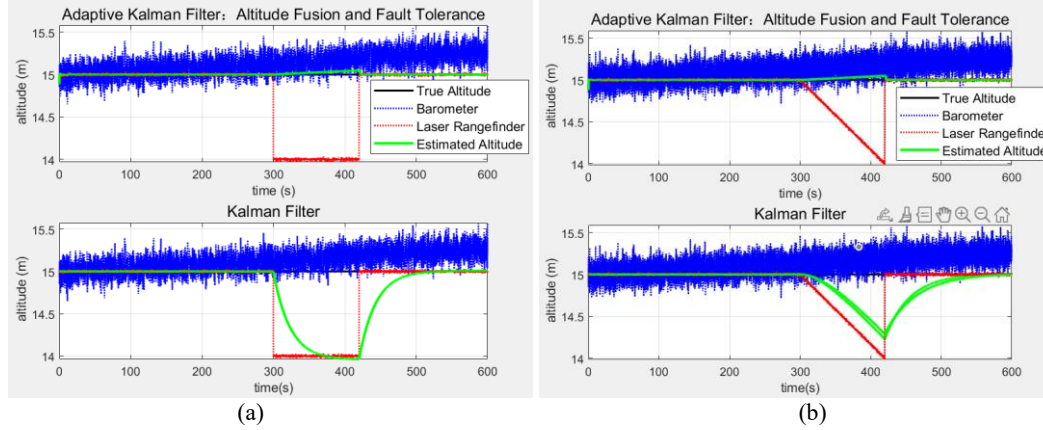


Figure 2. Laser step and ramp disturbance scenarios.

As can be seen from the figure, the non-adaptive fusion method fails to effectively suppress the influence of the laser disturbances, resulting in large deviations in the estimated altitude. In contrast, the proposed method, which combines an adaptive Kalman filter with barometer bias calibration, adaptively adjusts the measurement noise covariance R_{laser} in response to the residuals during the disturbance period. This adaptive mechanism gradually increases the variance of the laser measurement, reducing its weight in the data fusion process. As a result, the estimated altitude no longer follows the laser measurements closely during the disturbance, but instead, it gradually converges toward the barometer readings, thereby avoiding deviation caused by the laser step change. The final estimated altitude remains consistent with the barometer's trend, exhibiting stronger smoothness and stability and effectively mitigating the impact of anomalies on altitude estimation.

Furthermore, the bias state e remains stable during the disturbance period, preventing contamination of the barometer calibration process by abnormal laser measurements. This contributes to further improving the overall stability and robustness of the altitude estimation.

4.3 Estimation Results of the Bias Term

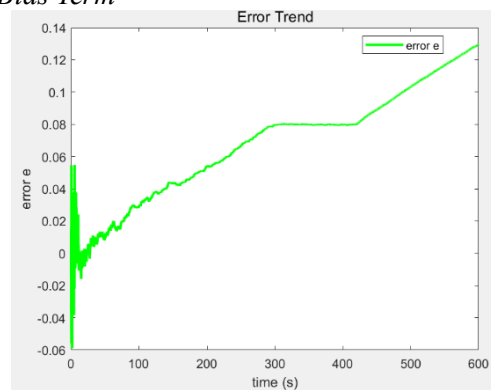


Figure 3. Error estimation result.

Figure 3 illustrates the variation of the bias term e of the barometer. It can be observed that when the barometer exhibits continuous drift, the bias term e can gradually track this drift trend, thereby achieving dynamic calibration of the barometer measurements. During the period of laser disturbances, e remains relatively stable and is not affected by abnormal laser measurements. Furthermore, after long-

term filtering, the convergence trend of e aligns well with the barometer drift, indicating that the introduction of e effectively improves the reliability of the fusion results.

4.4 Effect of Adaptive Parameter Adjustment

As shown in Figure 4, In both step-change and ramp-change scenarios, the comparative results demonstrate that when the laser residual increases, R_{laser} dynamically rises, effectively suppressing the influence of abnormal data on the estimated altitude. After the disturbance subsides, R_{laser} gradually decreases, allowing the laser information to regain weight in the fusion process.

Meanwhile, the process noise covariance Q moderately increases during the disturbance period, granting the altitude state greater freedom to vary and thereby maintaining a smooth transition in the estimation results.

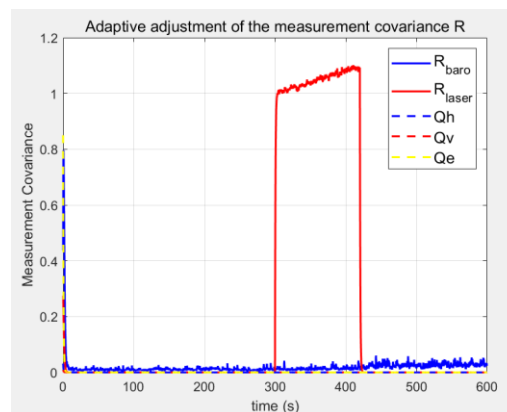


Figure 4. Covariance trend.

4.5 Analysis and Discussion

The simulation results confirm that the proposed adaptive Kalman filter-based multi-sensor height fusion and fault-tolerant method achieves robust and stable altitude estimation for indoor UAVs. By introducing a barometer bias state, the filter continuously compensates for barometric drift, preventing long-term offset in the fused altitude and validating the effectiveness of the state-augmented model. Under normal conditions, the estimated altitude closely follows the true value, balancing barometer and laser data to maintain both accuracy and smoothness.

During laser measurement anomalies, the adaptive mechanism dynamically increases the laser measurement noise covariance, reducing the influence of faulty data and allowing the estimate to converge back toward the barometer trend. This behavior is observed for both step and ramp disturbances, indicating good tolerance to abrupt and gradual errors. Analysis of the Kalman gain, residuals, and noise covariances shows that the filter adaptively re-weights sensor contributions in response to abnormal observations, maintaining robustness in multi-sensor environments.

However, as the process noise covariance converges to smaller values, the system becomes more prediction-driven, which may slightly degrade instantaneous tracking but improves overall stability—an advantage in indoor flight scenarios.

In recent years, machine learning has been increasingly applied to multi-sensor fusion and anomaly detection. Unsupervised learning can detect sudden laser or barometer anomalies early and adjust filter weights more rapidly, while deep learning can model sensor noise characteristics to improve filter accuracy and robustness. Future work will further explore these methods to enhance adaptability in complex indoor environments.

5 Conclusion

This paper proposes a multi-sensor altitude fusion and fault-tolerant method for indoor UAVs based on adaptive Kalman filtering. By introducing a barometer bias state and adaptively adjusting process and measurement noise covariances, the method continuously calibrates barometer drift and suppresses sudden laser errors. Simulation results show that:

1. The estimated altitude remains stable under slow barometer drift, eliminating cumulative bias.
2. During step or ramp anomalies in laser measurements, dynamic noise adjustment mitigates abnormal data and maintains smooth altitude estimation.
3. The adaptive variation of Kalman gain and noise covariances ensures robustness under varying measurement conditions.

In summary, the proposed method achieves accurate and fault-tolerant altitude estimation in complex indoor environments, but its response to anomalies remains limited. Future work will explore machine learning for faster fault detection and more precise adaptive tuning of Kalman filter parameters.

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